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Instability Suppression in a Swirl-Stabilized Combustor Using Microjet Air Injection

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In this study, we examine the effectiveness of microjet air injection as a means of suppressing thermoacoustic instabilities in a swirl-stabilized, lean-premixed propane/air combustor. High-speed stereo PIV measurements, taken to explore the mechanism of combustion instability, reveal that the inner recirculation zone plays a dominant role in the coupling of acoustics and heat release that leads to combustion instability. Six microjet injector configurations were designed to modify the inner and outer recirculation zones with the intent of decoupling the mechanism leading to instability. Microjets that injected air into the inner recirculation zone, swirling in the opposite sense to the primary swirl were effective in suppressing combustion instability, reducing the overall sound pressure level by up to 17 dB within a certain window of operating conditions. Stabilization was achieved near an equivalence ratio of 0.65, corresponding to the region where the combustor transitions from a 40 Hz instability mode to a 110 Hz instability mode. PIV measurements made of the stabilized flow revealed significant modification of the inner recirculation zone and substantial weakening of the outer recirculation zone.

I. Introduction

INCREASINGLY strict regulations on pollutant emission of gas turbines for power generation and aviation have led to the development of new concepts for combustor technology. Among them, lean-premixed (LPM) combustion has arisen as one of the most promising concepts for future gas turbine applications, due in large part, to the significant reduction in thermal NO_x production.[1] Combustion instability, manifested as discrete-frequency, large-amplitude pressure oscillations is a major challenge facing the design of practical LPM combustors. These instabilities are driven by feedback interactions between the heat release and acoustic fields. They arise due to the relatively low energy required to sustain them, and the absence of the attenuation mechanisms present in more conventional combustors operating in non-premixed mode.[1,2]

In order to achieve the reduction in pollutant emissions that lean-premixed combustion promises, effective means of creating stable operating regimes must be developed. Active control strategies, which attempt to either detect and correct, or preemptively correct for combustion instabilities have been studied, many of which focus on the modulation of fuel injection,[3,4] through small holes known as "microjets." Microjets allow for high injection velocities to be achieved with small mass flow rates.

Passive control strategies, if they can be developed, have the advantages of decreasing the complexity and increasing the reliability of instability suppression. The use of microjets has also been explored as a potentially simple means of introducing passive flow control to mitigate combustion instabilities. Microjet air injection has been shown to be effective in a lean-premixed backward-facing step combustor under certain conditions.[5-7]

The purpose of this study is to evaluate the potential effectiveness of microjet air injection for the suppression of thermoacoustic instabilities in an axisymmetric, swirl-stabilized combustor. Synchronized pressure data and high-speed images for stereo PIV were recorded over a range of lean equivalence ratios.

We designed and tested six different microjet injectors, each designed to modify the flow field in either the outer recirculation zone or the inner recirculation zone, with the intention of using the dynamics of the flow to decouple the feedback mechanism between the heat release of the flame and the acoustics.

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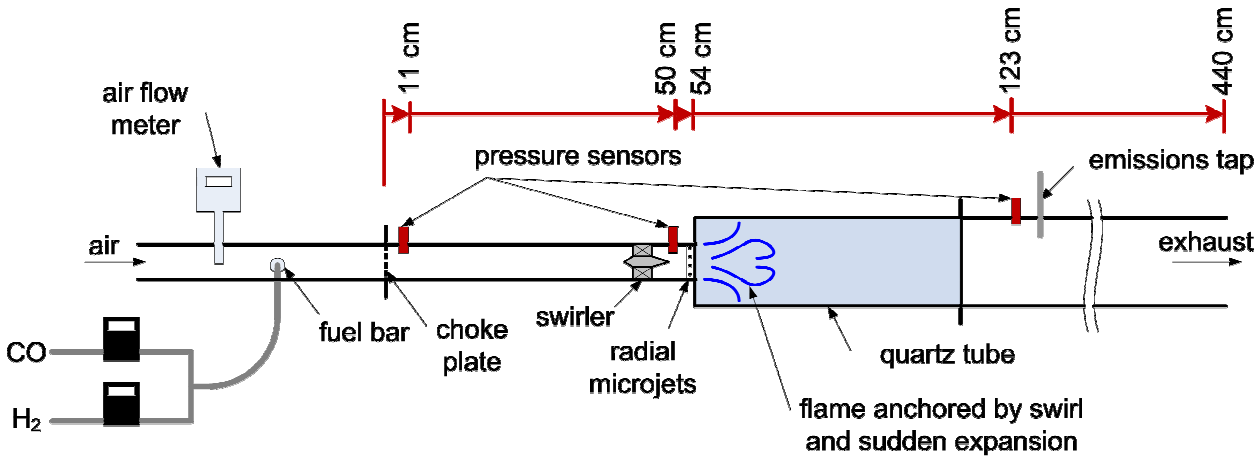

Figure 1. Schematic of the "swirl-counter-swirl" configuration, showing radial, counter-rotating microjets inducing swirl counter to the main air swirler.

II. Experimental Setup

A. Axisymmetric Swirl Combustor

The combustor consists of a cylindrical inlet pipe with a diameter of 38 mm, with a sudden expansion to 76 mm, as shown in Fig. 1. The inlet to the combustor is choked 54 cm upstream of the step. An axial swirler with eight vanes is located 10 cm upstream of the step and inducing clockwise swirl with a swirl number of 0.6. The length of the combustor and exhaust is 440 cm. Main air is supplied by an air compressor capable of delivering 110 g/s at 883 kPa. Fuel is supplied by Sierra C100M mass flow controllers. Separate fuel lines allow for propane or carbon monoxide to be supplied as the primary fuel and either can be enriched with hydrogen. Fuel is injected upstream of the choke plate.

B. Microjet Injector Design

We designed and tested six microjet injector configurations for this study. Injectors were classified as either "tangential" or "radial", and as either "straight," "co-rotating," or "counter-rotating." Tangential microjets inject air tangential to the cylindrical surface of the combustor, whereas radial microjets inject air in the plane perpendicular to the longitudinal axis of the combustor.

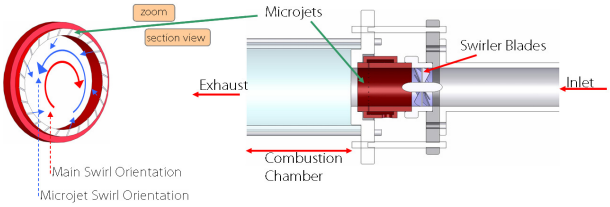

Figure 2. Schematic of the "swirl-counter-swirl" configuration, showing normal, counter-rotating microjets inducing swirl counter to the main air swirler.

The designs classified as counter and co-rotating introduce swirl to the injected air in the same sense as that of the main air for the co-rotating cases, and in the opposite sense as the main air for the counter-rotating cases. The straight tangential microjets inject air parallel to the longitudinal axis of the combustor, and the straight radial microjets inject air along radii of the combustor. In Fig. 2, we show a schematic of radial, counter-rotating microjets.

The tangential co-rotating and counter-rotating microjets were designed to provide a swirl number of about 0.6, the same as the swirl number of the main swirler. This was achieved by angling the microjets 35 degrees from longitudinal axis of the combustor. The same relative angle of the microjets was used for the co-rotating and counter-rotating, radial microjets, which are offset from the radial direction by 35 degrees. No attempt has yet been made to optimize the offset angle for any of the four swirl-inducing microjets.

C. Pressure Diagnostics and Stereo PIV

Diagnostics were conducted using pressure microphones located at several positions along the length of the combustor, and a high-speed stereo particle image velocimetry (SPIV) system was setup to allow for further investigation into the physics of the flow in conditions of particular interest.

Pressure was measured using Kulite MIC-series microphones, offset from the combustor in semi-infinite line configurations and Kistler 7061B water-cooled pressure sensors, flush-mounted with the inside of the combustor. Pressure data was recorded at 10 kHz, and was synchronized with the high-speed cameras for the PIV tests.

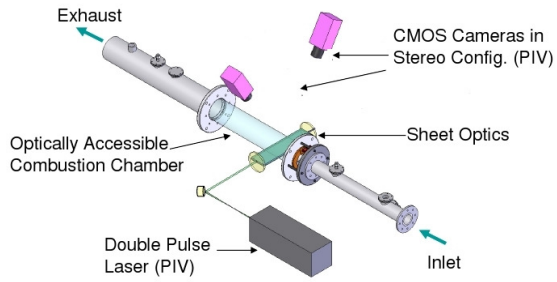

Figure 3. Schematic of the stereo PIV setup.

Two high-speed CMOS cameras were mounted above the combustor, as shown in Fig. 3, to capture images at a rate of 2 kHz. A 50 W Nd:YLF laser was operated in twin pulse mode, and the beam was routed through sheet forming optics to produce a sub-millimeter light sheet. The field of view was a 76 mm by 76 mm square spanning the entire diameter of the combustor and the region immediately downstream of the expansion plane.

Aluminum oxide particles of 0.9 to 2.2 μm diameter were injected into the flow as seeding particles. Air downstream of the choke plate was diverted into a cyclone-type seeder and back into the inlet flow upstream of the swirler. The inner surface of the quartz tube, below the light sheet, was sandblasted to reduce specular reflections. DaVis 7.2 software provided by LaVision processed the PIV images.

Each individual test lasted 0.55 s during which 2200 images were recorded, creating 1100 velocity fields. For stable cases, a simple time-average of all velocity fields was computed. For unstable cases, velocity fields were averaged based on the corresponding phase of the upstream pressure oscillations. One cycle of the pressure oscillation was divided into 80 equally spaced bins, and velocity fields were binned accordingly, and all velocity fields in a given bin were averaged. Data presented here for the unstable cases represent two to three separate trial runs, with an average of 25 to 40 velocity fields per bin.

III. Results

A. Swirl Combustor Stability

We conducted a series of experiments to explore the impact that each microjet injector design had on the stability of the combustor. For each injector, we set the Reynolds number at 19000, corresponding to a mass flow rate of 8.9 to 9.7 g/s and a mean inlet velocity of 7.1 to 7.5 m/s, depending on the inlet equivalence ratio. We fixed the mass flow rate through the microjet injectors at values from 1.0 to 3.0 g/s, and varied the inlet equivalence ratio from 0.85, as measured at the choke plate, down to the lean blow-off limit.

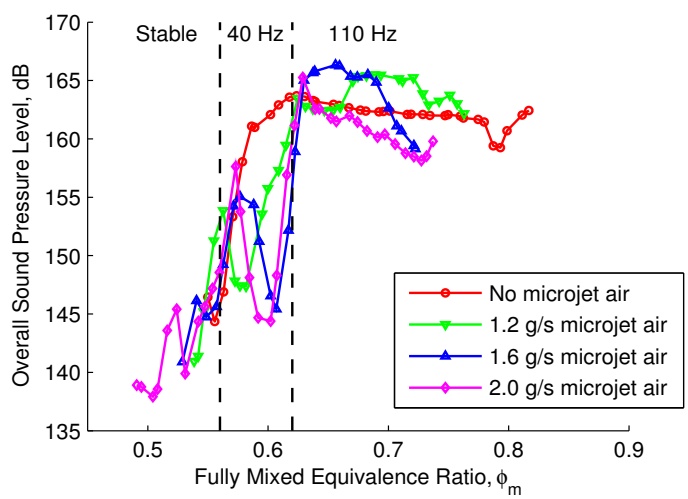

Figure 4. Overall sound pressure level as a function of the fully mixed equivalence ratio for the SCS configuration.

Additionally, at several fixed values of the inlet equivalence ratio, just above the onset of the 110 Hz unstable mode of the combustor, we varied the microjet flow rate from 1.0 to 4.0 g/s. The radial, counter-rotating microjet injector was the only configuration that showed promise of a significant reduction in the overall sound pressure level over the baseline case with no microjet air injection. This configuration, labeled the "swirl-counter-swirl" (SCS) configuration, was chosen for further study.

For the SCS configuration, the inlet equivalence ratio was varied from an inlet equivalence ratio of 0.85, down to the lean blow-off limit at fixed microjet flow rates of 1.2 g/s, 1.6 g/s, and 2.0 g/s. In all cases the flow at

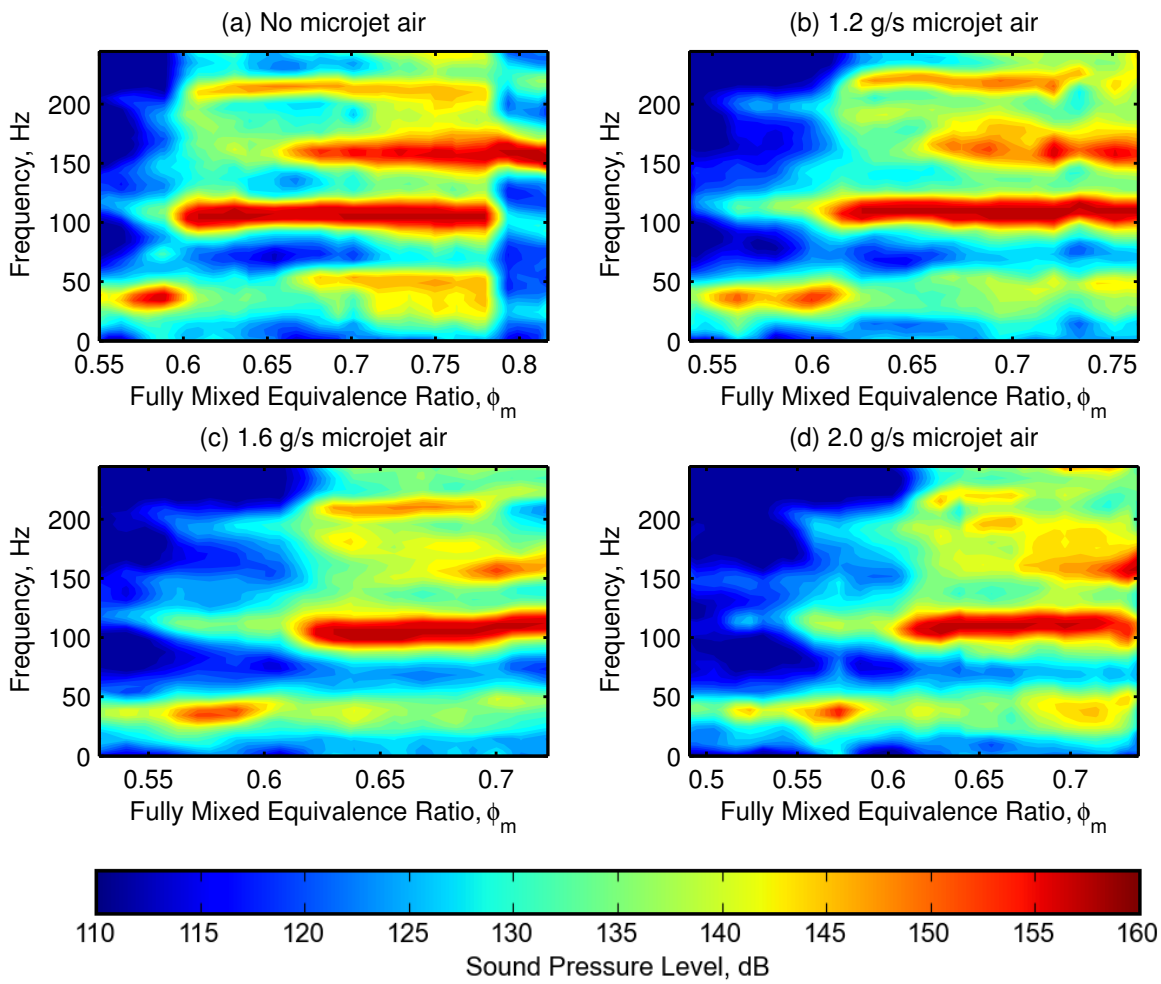

Figure 5. Sound pressure level spectra for a range of microjet flow rates in the SCS configuration.

the microjets was choked, preventing microjet flow rates from coupling with pressure oscillations in the combustor. In Fig. 4, we present the overall sound pressure level for the baseline and three microjet flow rates as a function of the mixed equivalence ratio. The mixed equivalence ratio is computed assuming that the inlet fuel-air mixture and the secondary air from the microjets mix perfectly before burning.

The overall sound pressure level as a function of both frequency and the mixed equivalence ratio is shown in Fig. 5 for the combustor with no microjet injection, 1.2 g/s, 1.6 g/s, and 2.0 g/s. In the baseline case, the combustor transitions from a 40 Hz to a 110 Hz mode, with no significant reduction in OASPL during the transition. At an equivalence ratio of 0.56, the flame becomes stable.

The stable flame is not compact, and extends downstream into the exhaust; the flame length is on the order of meters. The two unstable modes we studied, and the microjet stabilized mode all produce compact flames, which are completely contained within the quartz combustion chamber; the flame length for these modes is less than 0.5 m.

B. Flow Field Observations without Microjet Injection

We conducted two-dimensional and three-dimensional PIV measurements on the first three operating modes of the swirl combustor: the stable mode, the 40 Hz instability, and the 110 Hz instability. The stable mode of the combustor, which is only observed at low equivalence ratios, produces a long, conical flame that extents from the bluff body of the swirler downstream of the main combustion chamber. The time-averaged axial and tangential velocity profiles of this flame are shown in Fig. 6. The important flow features in the stable flame are the large inner recirculation zone, the outer recirculation zone and the relatively strong jet emanating from the edge of the inlet pipe. The tangential velocity field indicates that the flow is swirling clockwise as viewed from the inlet direction, which is the same sense of swirl as the main air swirler.

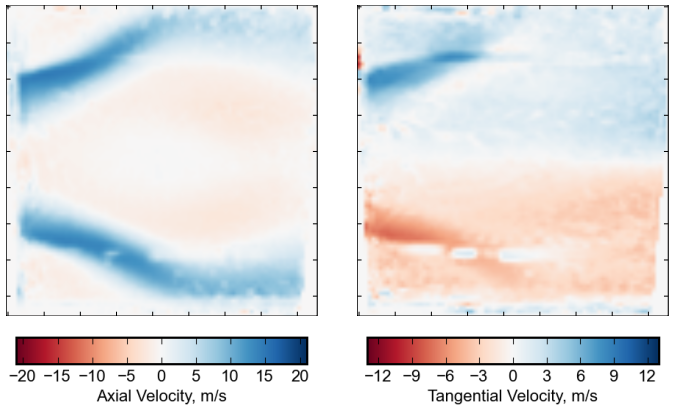

Figure 6. Axial (left) and tangential (right) velocity contours for the stable flame with no microjet air injection at an equivalence ratio of 0.70.[*] Flow is from left to right.

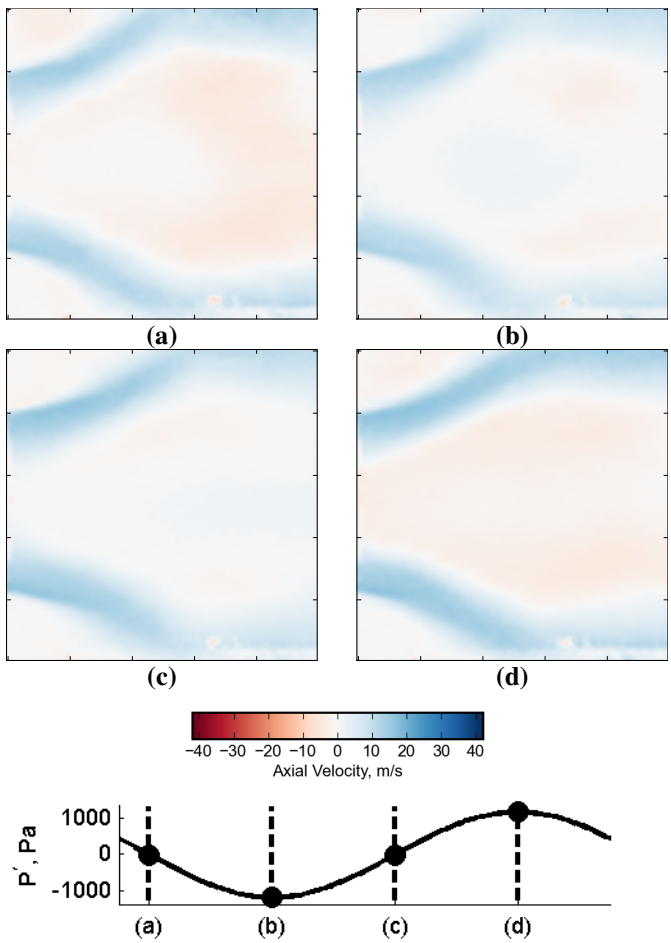

Figure 7. Axial velocity contours for the 40 Hz mode instability with no microjet air flow at an equivalence ratio of 0.68[*] as a function of the upstream pressure fluctuation, P´. Flow is from left to right.

[*] The equivalence ratios seen in Fig. 6 and Fig. 7 do not compare to those shown in Fig. 5 for the stable and 40 Hz modes due to a strong hysteresis effect, and the equivalence ratio at which the combustor was ignited. Investigation of microjet suppression called for the careful control of the startup procedure prior to recording data, but the procedure was not necessary to characterize the basic modes of the combustor without microjets.

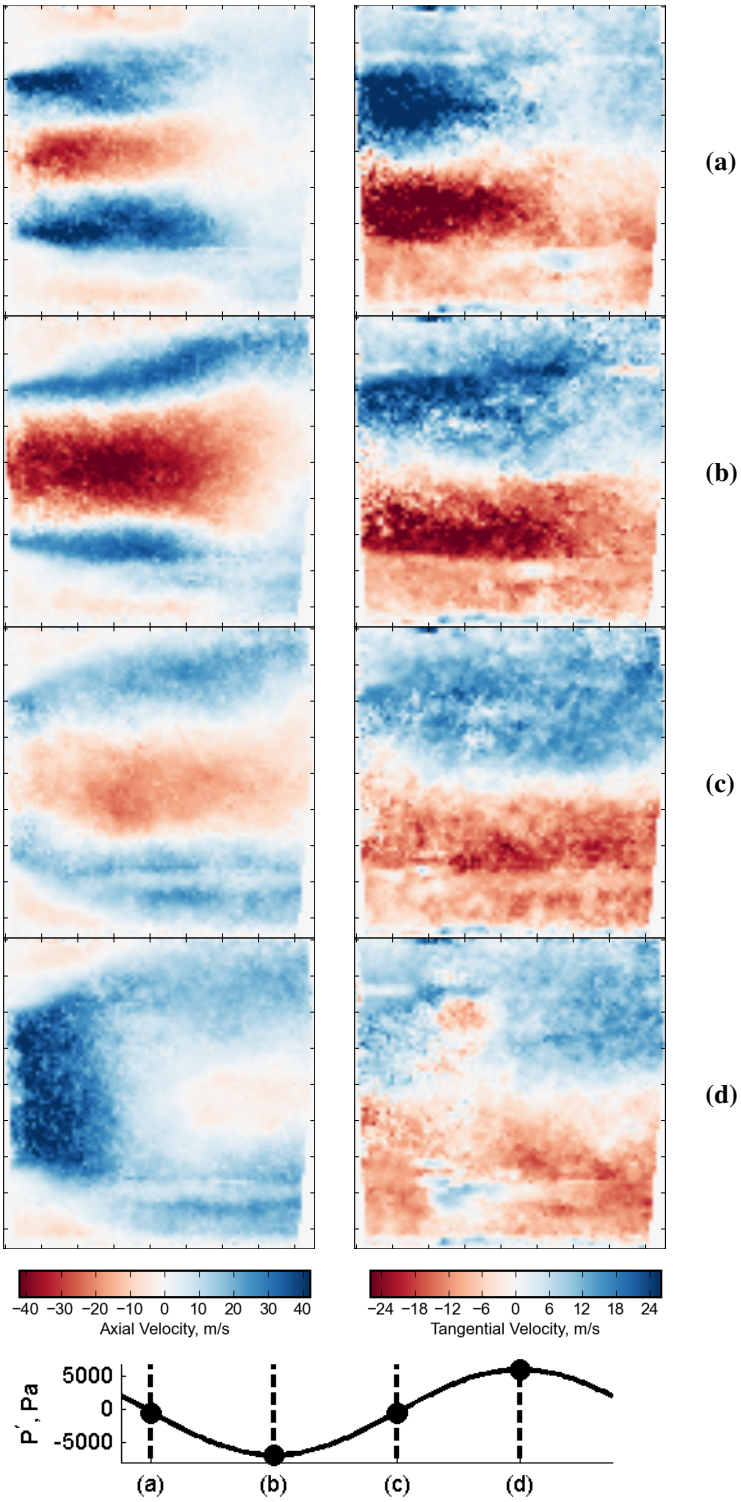

Figure 8. Axial (left) and tangential (right) velocity contours for the 110 Hz mode instability with no microjet air flow at an equivalence ratio of 0.75 as a function of the upstream pressure fluctuation, P´. Flow is from left to right.

In Fig. 7, we show phase-averaged axial velocity field for the 40 Hz mode at the minimum, maximum, and zeros of its oscillation cycle. This mode is characterized by the relative steadiness of the jet and weak oscillations in the magnitude of the inner recirculation zone. The outer recirculation zone remains largely unaffected by the oscillations. The oscillations are associated with the shedding of large vortex rings within the inner recirculation zone. The 40 Hz mode has a broad peak when compared with other modes of instability in this combustor, and the result is that the peak-to-peak amplitude of the dominant frequency of oscillation is small compared to that of other modes.

We show a sequence of axial and tangential velocity contours at different points in the cycle for the 110 Hz instability in Fig. 8. Data were phase-averaged, and the results are plotted for the maximum, minimum and zeros of the upstream pressure fluctuation. The 110 Hz mode features a sharper peak than the 40 Hz mode, and the resulting peak-to-peak amplitude of the dominant frequency of pressure fluctuations is correspondingly larger than that of the 40 Hz mode.

The 110 Hz mode shows strong dynamic behavior. We observe periodic collapse of the inner recirculation zone throughout the cycle. A strong recirculation zone forms with axial jets; as the recirculation zone weakens, the jets weaken and are deflected outward toward the walls of the combustion chamber; finally, the recirculation zone completely gives way to a strong forward jet.

The largest forward and reverse velocity magnitudes are seen in the 110 Hz mode, with the jets reaching a velocity of 40 m/s, and the strong inner recirculation zone reaching a velocity of 30 m/s.

Characteristic of the tangential velocity in all three cases—the stable mode, the 40 Hz instability, and the 110 Hz instability—is the observation that the largest tangential velocities correspond to the presence of strong jets from the inlet pipe surrounding the inner recirculation zone.

C. Flow Field Observations with Microjet Injection in SCS Configuration

Time-averaged contours of the axial and tangential velocity for the case of swirl stabilized flow are shown in Fig. 9. There are several features that distinguish the microjet stabilized flow field from the stable flow field of the combustor. The most notable feature is that the tangential velocity of the stabilized flow is counterclockwise, as viewed from the inlet, looking downstream. That is, the sense of swirl in the flow is dominated by the induced swirl of the microjets, not the main air swirler. The axial velocity reveals the presence of a bowl-shaped inner recirculation zone and an outer recirculation zone that is significantly diminished in both intensity and extent.

The swirl number of the stabilized flow is 0.9, in the opposite sense as the stable flame. Whereas in all other cases we examined, the largest tangential velocities corresponded to the presence of strong jets (relative to the rest of the flow field), there are no strong jets in the stabilized flame, and the largest tangential velocities appear near the longitudinal axis of the combustor.

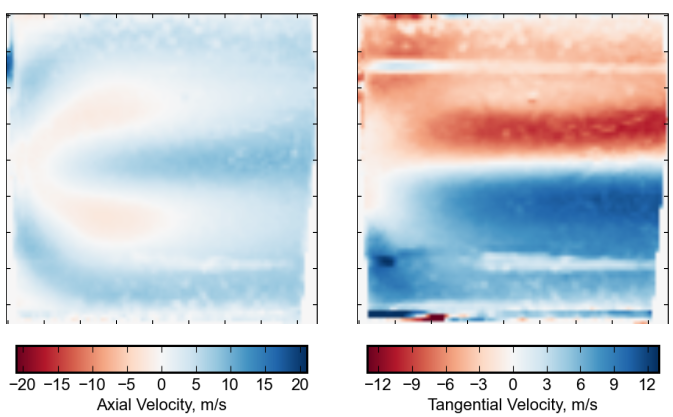

Figure 9. Axial (left) and tangential (right) velocity contours for the microjet stabilized flame with 1.6 g/s of microjet air injection at a fully mixed equivalence ratio of 0.63. Flow is from left to right.

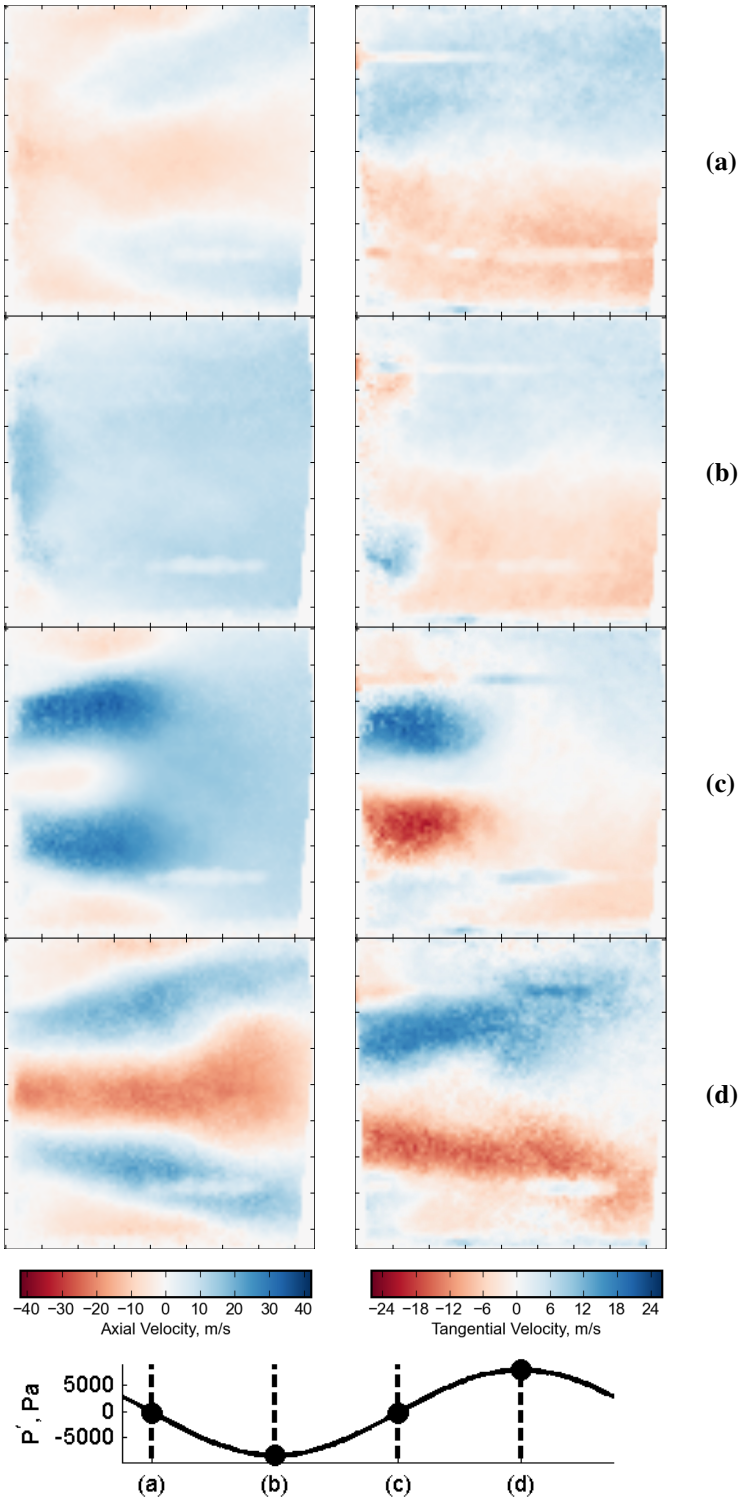

Figure 10. Axial (left) and tangential (right) velocity contours for the 110 Hz mode instability at a mixed equivalence ratio of 0.68 with 0.9 g/s of microjet air flow as a function of the upstream pressure fluctuation, P´. Flow is from left to right.

We began to investigate the effect of the SCS microjet configuration in unstable cases, where the air flow rate through the injectors is insufficient to stabilize the flame. In Fig. 10, we show the a sequence of phase-averaged axial and tangential velocity contours for a case in which the 110 Hz instability mode was active and microjet air was being injected at a flow rate of 0.9 g/s. The inlet equivalence ratio was 0.75, and the fully mixed equivalence ratio was 0.68.

The dynamics of the axial velocity are similar to those of the 110 Hz instability without microjet injection, with several notable exceptions. The collapse of the inner recirculation zone is phase-shifted when compared to the collapse with no microjet air injection. Additionally, the tangential velocity is more active than all other observed modes of operation. The bulk swirl is clockwise, dominated by the main air swirler, but a small zone of counter-clockwise swirl, dominated by the microjet injection, appears when the axial velocity is low, as shown in Fig. 10(b). Strong jets form, as seen in Fig. 10(c) and 10(d), and we observe correspondingly large tangential velocities in the clockwise direction, displacing the counterclockwise zone into the outer recirculation zone.

IV. Conclusion

In this paper, we have used high-speed stereo PIV, synchronized with pressure measurements, to study the flow structure of thermoacoustic instabilities in a swirl-stabilized, lean-premixed propane/air combustor. We observed that the inner recirculation zone, formed downstream of the swirler, plays a dominant role in the coupling that occurs between acoustic oscillations and heat release that leads to combustion instability.

We developed six sets of microjet air injectors, intended to break the feedback mechanism that give rise to instabilities by changing the dynamics of the flow field in the combustor. Consistent with our findings regarding the dominant role of the inner recirculation zone, we were able to achieve a 17 dB reduction in the overall sound pressure level by injecting high-velocity air into the inner recirculation zone. We observed the creation of a gap between the 40 Hz and 110 Hz instabilities of the swirl combustor when the microjets injected air swirling in the opposite sense of the inlet air, using the swirl-counter-swirl microjet configuration.

Acknowledgments

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References

[1] Huang, Y., and Yang, V., "Dynamics and Stability of Lean-Premixed Swirl-Stabilized Combustion," *Progress in Energy and Combustion Science*, Vol. 35, 2009, pp. 293-364.

[2] Keller, J. J., "Thermoacoustic Oscillations in Combustion Chambers of Gas Turbines," *AIAA Journal*, Vol. 33, No. 12, 1995, pp. 2280-2287.

[3] Lee, J. G., Kim, K., and Santavicca, D. A., "Effect of injection Location on the Effectiveness of an Active Control System Using Secondary Fuel Injection," *Proceedings of the Combustion Institute*, Vol. 28, 2000, pp. 739-746.

[4] Tachibana, S., Zimmer, L., and Kurosawa, Y., "Active Control of Combustion Oscillations in a Lean Premixed Gas-Turbine Combustor," International Journal of Vehicle Design, Vol. 43, Nos. 1-4, 2007.

[5] Hudgins, D. E., "Suppression of Premixed Combustion Dynamics Utilizing Microjet Air Injection," M. S. Thesis, Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA, 2008.

[6] Altay, H. M., "Physics-Based Flame Dynamics Modeling and Thermoacoustic Instability Mitigation," Ph. D. Thesis, Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA, 2009.

[7] Altay, H. M., Speth, R. L., Snarheim, D., Hudgins, D. E., Ghoniem, A. F., Annaswamy, A. M., "Impact of Microjet Actuation of a Backward-Facing Step Combustor," 45th AIAA Aerospace Sciences Meeting and Exhibit, AIAA, Reno, NV, 2007, AIAA Paper 2007-563.